

Machine Learning Tutorial

Understanding Random Forest Classification: Predicting Breast Cancer Diagnosis and
Explaining Feature Importance

Github: <https://github.com/ammarikram7867-cpu/Machine-Learning-Tutorial>

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Introduction

Machine learning has become a life-saver in resolving classification challenges across a wide variety of domains, including healthcare, finance, marketing, and more. The number of algorithms is abundant, with one of the most popular being Random Forest, which delivers strong predictive performance and is relatively easy to use. It can handle large datasets, a wide variety of input variables, and identify the most significant variables for prediction. In this tutorial, we will see how a Random Forest Classifier works by applying it to the Breast Cancer Wisconsin dataset in scikit-learn. The goal is not just to create a predictive model but to understand "how" and "why" the algorithm made the decision and how to interpret the result. At the end of this tutorial, you will be able to train an RF model, evaluate its performance, and interpret the results using feature importance techniques.

Learning Objectives

After completing this tutorial, you should be able to:

- Fundamental concepts behind Random Forest Classification.
- To grasp the reasons for the success of ensemble methods.
- Write a Python scikit-learn implementation of a Random Forest of classifiers.
- Compare the classification performance across several criteria.
- Understand the limits of using feature importance.
- Use the same process to design and build your own classification problems.

Introducing the Breast Cancer Wisconsin Dataset

To demonstrate Random Forest Classification, we will use the Breast Cancer Wisconsin dataset from scikit-learn. The data are computed on digitised images of breast tissue samples. The data are organised in a table, with a single row per tumour sample and a single column for each numeric feature describing cell-nucleus properties, including radius, texture, perimeter, and smoothness (Breiman, 1999). The target variable has two classes:

- **Malignant (0):** Cancerous tumours.
- **Benign (1):** Non-cancerous tumours.

The dataset contains 569 observations and 30 numerical features, making it an excellent example for binary classification.

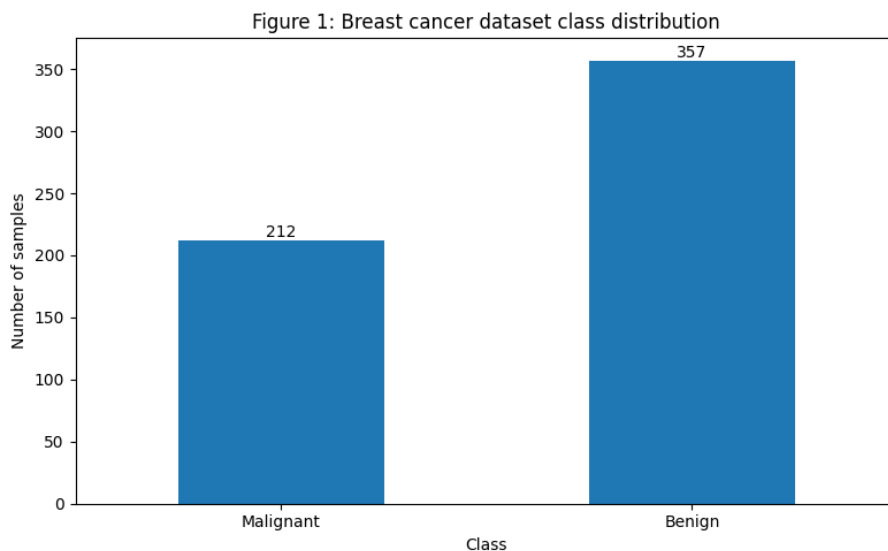


Figure 1: Dataset Class Distribution

The first visualisation in the tutorial should be a bar chart showing the counts of malignant and benign samples. When you are ready to train any Machine Learning model, you need to examine the class distribution first, as imbalanced datasets can lead to errors in your model for the majority class in your data. The advantage here is the reasonably balanced nature of the Breast Cancer dataset, which will permit us to train and test the classifier efficiently.

From One Decision Tree to a Forest

To understand Random Forests, you need to understand what decision trees are first. A supervised learning algorithm that recursively makes predictions by splitting the data based on feature values is called a decision tree. The algorithm selects the feature which maximally separates the classes at each node (Breiman, 2001). Additionally, the process will reach a leaf node that yields the final prediction. A tree can, for instance, inquire if the mean radius of a tumour is above a certain value. Based on the response, it can then examine a different characteristic, such as texture or perimeter, before deciding on a classification.

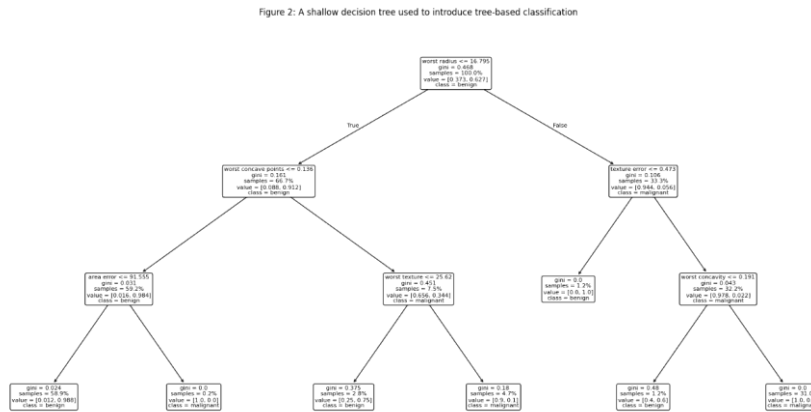


Figure 2: Simple Decision Tree Diagram

A shallow decision tree diagram should illustrate the structure of a tree, including:

- Root node
- Internal decision nodes
- Leaf nodes containing predictions

Decision trees are simple to understand and visualise and can be used for many applications. The problem with them, however, is that they can overfit the training set. Overfitting is the ability to learn not only the underlying patterns but also the noise in the training set. A tree that is "overfitted" can be very good at the training set, but not at test sets. The limitation has inspired the development of ensemble techniques such as Random Forest (Muduli *et al.*, 2023).

How Random Forest Works

Random Forest is an ensemble learning algorithm that combines many decision trees to build a predictive model. Rather than having a single potentially unreliable tree, the algorithm creates an ensemble of trees and averages their forecasts (Parmar, Katariya and Patel, 2019). The process can be broken down into five steps:

Step 1: Create Bootstrap Samples: The algorithm samples the data numerous additional times from the training data set, using sampling with replacement. These are named "bootstrap samples".

Step 2: Train Multiple Decision Trees: Train one decision tree for each of the n bootstrap samples. Draw a bootstrap sample for each sample to train one decision tree.

Step 3: Random Feature Selection: Only a subset of features is considered at each division of a tree. This adds some variety to the trees and helps keep them from becoming too alike.

Step 4: Generate Predictions: These trees independently predict malignancy vs benignity.

Step 5: Majority Voting: The final prediction is the one that wins the most votes. The class with the most votes will be the model's prediction.

Figure 3: Random Forest workflow: bootstrap samples → multiple trees → majority vote

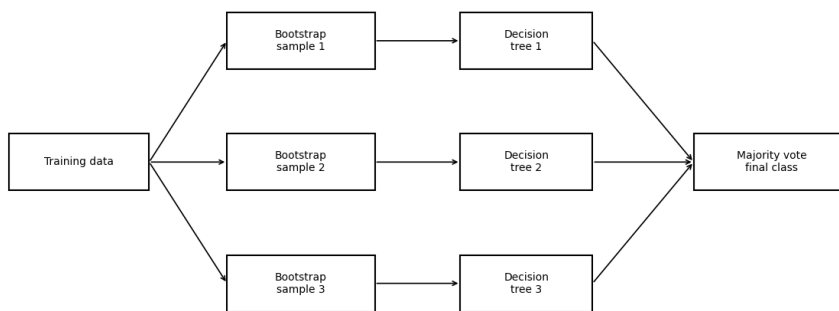


Figure 3: Random Forest Workflow Diagram

The workflow diagram representation makes it easier for learners to comprehend that the Random Forest is simply a group of individual decision trees collaborating.

Why Random Forest Improves Performance

The power of Random Forest lies in the idea that many weak or noisy models can combine to yield a powerful and stable prediction. Decision trees may be incorrect in some instances due to chance, arising from the entire data set being split into smaller subsets for training each tree. Some errors may be due to chance when a decision tree is trained because the data set is divided into smaller subsets. It may be the case, however, that errors across different trees become independent of one another, and that consensus prediction may actually yield higher overall accuracy (Sulistiani, Wulandari and Astuti, 2022). Two key concepts describe this improvement:

Diversity: There are different training examples and feature subsets for each tree. Therefore, the trees make different mistakes.

Aggregation: The intuition here is that the error from majority-voting predictions will be reduced, so generalisation will improve.

Figure 4: Majority voting example using predictions from nine trees

Actual label for this test sample: Benign

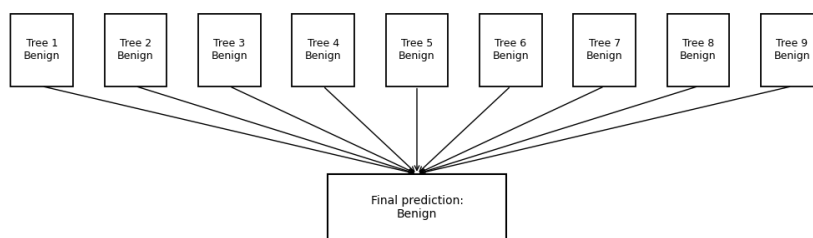


Figure 4: Majority Voting Example

A helpful illustration is a chart that illustrates several trees casting votes for malignant or benign. The class makes the final prediction based on the majority of votes. This fact illustrates the advantages of using ensemble models compared to single models.

Implementing Random Forest in Python

The practical implementation of Random Forest Classification involves a typical machine learning process. First, import the necessary libraries:

- NumPy
- Pandas
- Matplotlib
- Seaborn
- Scikit-learn

Next, load the Breast Cancer dataset using:

```
from sklearn.datasets import load_breast_cancer
```

The data set is then split into 'train' and 'test' sets with the function `train_test_split()`. The training set contains data used to build the model, and the testing set contains data used to test the model's generalisation capabilities. It can be used to create the Random Forest classifier:

```
from sklearn.ensemble import RandomForestClassifier
```

After training the model with the `fit()` method, predictions are generated using the `predict()` method.

Figure 5: Train-test modeling pipeline

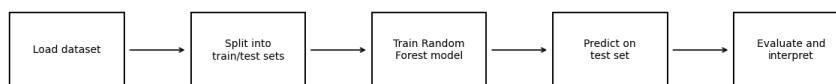


Figure 5: Train-Test Workflow

This workflow applies to almost every supervised learning problem and provides a useful template for future projects.

Evaluating the Model

Creating a model is just the first step. We shall also have to ascertain its efficiency. It is possible to define many evaluation metrics for a classification problem.

Accuracy: To measure how many observations are classified correctly, the percentage is used as this measure. While accuracy is intuitive, it can also be confusing, particularly when classes are imbalanced (Li *et al.*, 2023).

Precision: Precision is the ratio of the number of correct positive predictions to the total number of predictions. In medical use, precision is the number of tumours correctly classified as malignant.

Recall: Recall indicates the proportion of people who have the true case that are correctly identified. In the medical field, remembering information is important, as failing to recognise a malignant tumour can have grave consequences.

F1-Score: F1-Score is a score that merges precision and recall and hence is useful where both types of error are significant (Sulistiani, Wulandari and Astuti, 2022).

Figure 6: Confusion matrix for Random Forest predictions

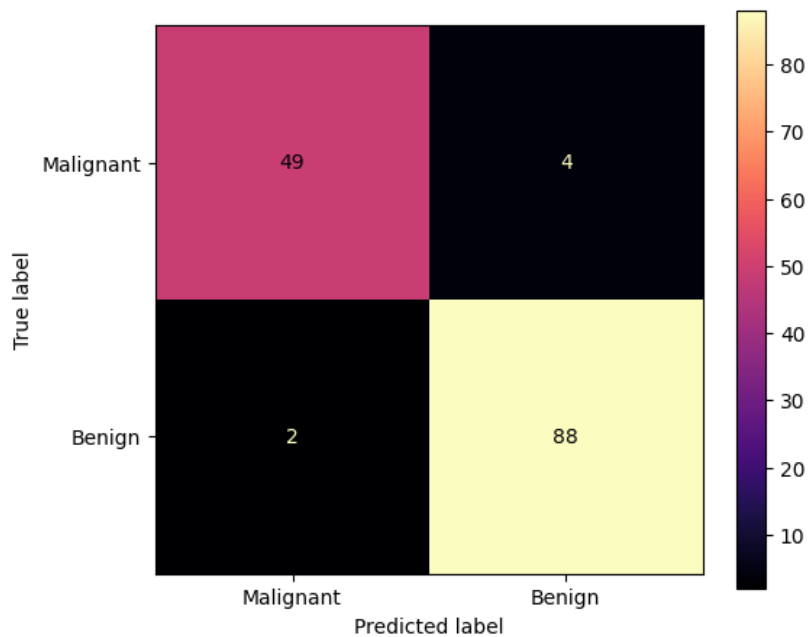


Figure 6: Confusion Matrix

The confusion matrix provides a detailed breakdown of predictions:

- True Positives
- True Negatives
- False Positives
- False Negatives

A false negative in the diagnosis of breast cancer is particularly problematic, as this indicates a tumour that was found to be malignant by the physician of the model but was misdiagnosed as benign. These errors can be easily understood by using a heatmap visualisation.

Figure 7: ROC curve for Random Forest classifier (AUC = 0.995)

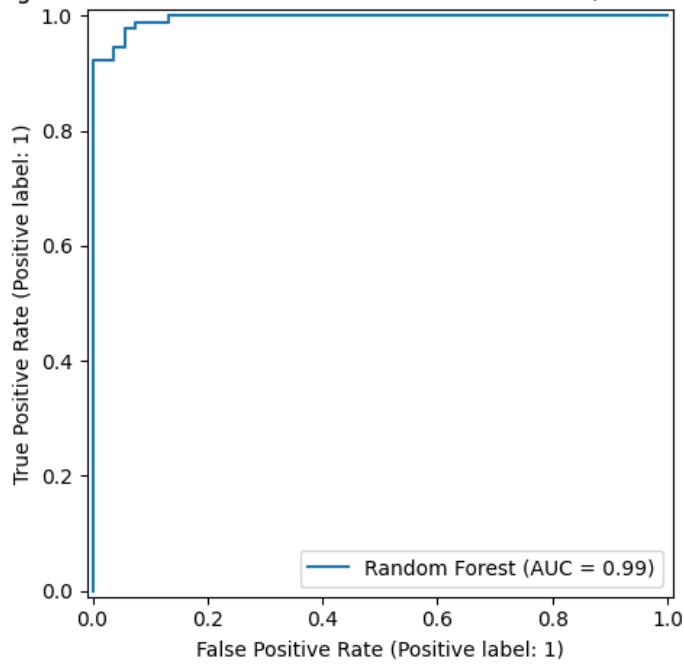


Figure 7: ROC Curve

The receiver operating characteristic (ROC) curve is a plot of sensitivity (true positive rate) against specificity (false positive rate). The Area Under the Curve (AUC) is a summary measure of the model's ability to distinguish between datasets. The curve closest to the top-left corner indicates the best classification (Wang, 2022).

Understanding Feature Importance

Random Forest offers a significant advantage for estimating feature importance. Learning about feature importance addresses a crucial question:

- Which variables contributed most to the model's predictions?

Understanding the structure and importance of all features may help gain insights and increase trust in machine learning models.

Impurity-Based Feature Importance

Random Forest computes feature importance using the mean of the tree contribution scores of all the trees. Consistent features in informative splits are assigned higher importance scores (Minnoor and Baths, 2023).

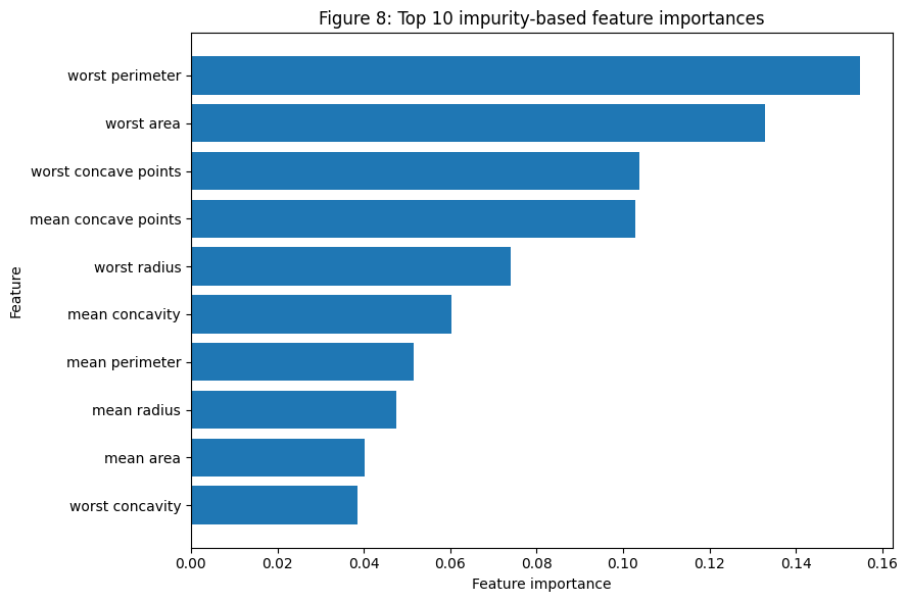


Figure 8: Feature Importance Bar Chart

A bar graph of the top 10 most important features can be used to identify which measures are most important for predicting breast cancer diagnosis. Some of these features, such as mean radius, worst perimeter, or worst concave points, are often the most significant predictors. Impurity-based importance has a problem, however. It may be useful only for variables with a large number of distinct values and may not always be useful for explaining (Minnoor and Baths, 2023).

Permutation Importance

Permutation importance offers an alternative approach. The idea is simple:

1. Measure the model's original performance.
2. Randomly shuffle one feature.
3. Measure the new performance.
4. Calculate the performance drop.

If shuffling a feature causes a large decrease in performance, that feature is considered important.

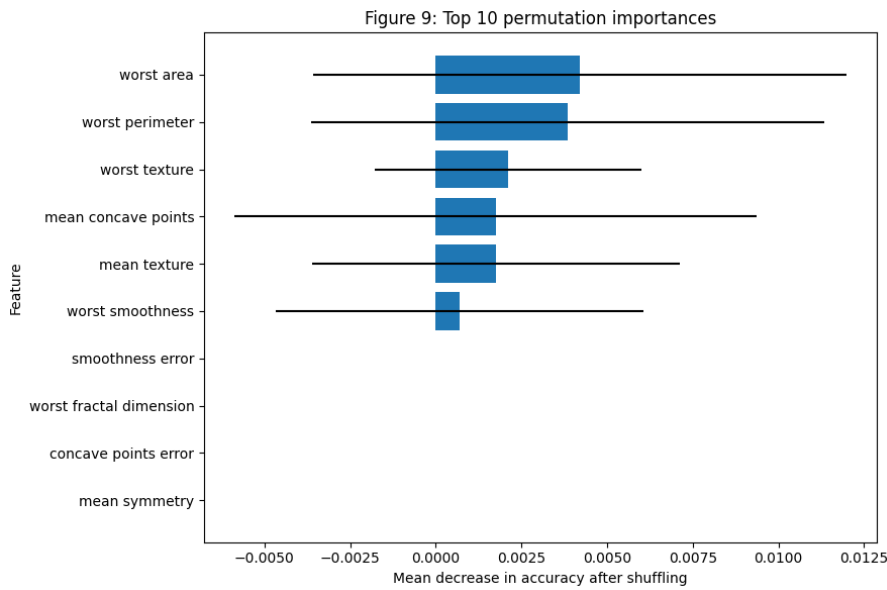


Figure 9: Permutation Importance Comparison

This should compare with the importance scores obtained from randomly shuffled feature importance scores. It can also be easier to interpret permutation importance, as it directly reflects the model's dependence on that particular variable (Wang, 2022).

Limitations and Responsible Interpretation

Random Forest is a strong algorithm, but it is not without flaws. First, this is less interpretable than a single decision tree. (Feature importance gives some explanation; the decision-making process used in hundreds of trees can be hard to understand). Secondly, the feature importance measures can be biased, especially for highly correlated features. Thirdly, good prediction does not necessarily mean causation. A feature can be very predictive, but it does not actually cause the result (Muduli *et al.*, 2023). Lastly, machine learning models should not replace expert judgment, particularly in critical fields like healthcare. Rather, they are meant to serve as decision-support mechanisms that help professionals make informed decisions.

Concluding Remarks

Random Forest Classification (RFC) is one of the most effective and widely used machine learning algorithms for classification problems. It can be used to make accurate predictions by combining many trees obtained via bootstrapping and voting, thereby preventing the model from overfitting. In this tutorial, we learned all about implementing the random forest model using the Breast Cancer Wisconsin dataset. The first step was to get an understanding of the data and the drawbacks of a single decision tree. We then learned how the Random Forest algorithm leverages the formation of a forest of trees, its advantages, and how to build it in Python. Evaluating accuracy, precision, recall, F1-score, confusion matrices, and ROC curves to assess classification performance was also explored. Finally, we looked into techniques for explaining the features that affect the model's predictions. Overall, the point is that building a successful machine learning model isn't just about training a single model. It involves an awareness of the data, a careful assessment of performance, careful interpretation of the results, and clear communication of limitations. With that said, you will feel confident that you can use Random Forest Classification and feature importance methods with your own datasets and real-life machine learning projects.

References

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